

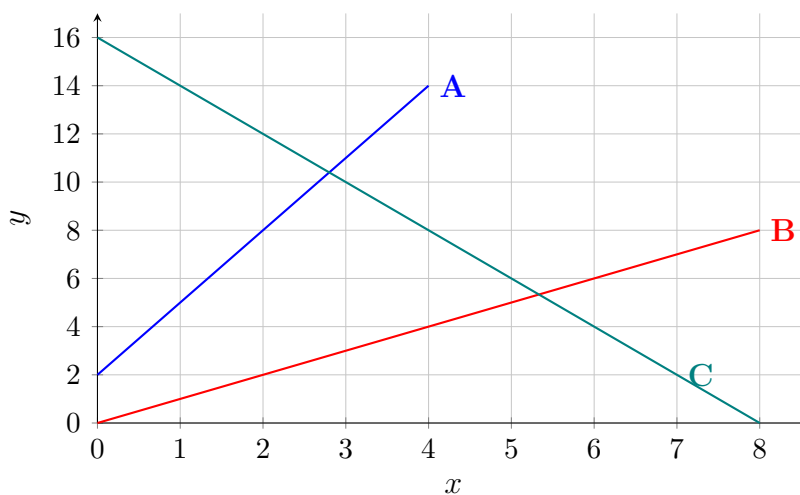
Slope and Rate of Change

Slope from two points, rate of change read from a table, rate of change read from an equation, and comparing two rates given in different forms

Before you start.

- Slope is a *rate*: how much y changes for each 1 that x changes. $m = \frac{y_2 - y_1}{x_2 - x_1}$
- From a **graph**, pick two points the line passes exactly through. From a **table**, use any two rows. From an **equation** in the form $y = mx + b$, the rate is m .
- To compare two rates given in different forms, get both into the same units first, then compare the numbers.
- A negative rate means the quantity is going *down*. When you compare rates that include negatives, order them on a number line, not by size.

Figure 1. *Used by Problems 2, 4 and 7.* Three lines, each passing exactly through lattice points of the grid.



1. Find the slope of the line through (1, 2) and (4, 8).

Answer: _____

2. Use **Figure 1**. Line **B** passes through the origin and through $(8, 8)$. Find its slope.

Answer: _____

3. A taxi charges a flat pick-up fee plus a fixed amount per mile. For a ride of d miles the cost in dollars is $C = 4d + 7$.

(a) What is the rate of change, in dollars per mile?

Answer: _____

(b) What does a 3 mile ride cost, in dollars?

Answer: _____

4. Use **Figure 1**. Line **A** passes through $(0, 2)$ and $(4, 14)$. Find its rate of change.

Answer: _____

5. Water runs into a tub at a steady rate. The table shows how many litres are in the tub after x minutes.

x (minutes)	2	5	8
litres in tub	14	32	50

- (a) How many litres run in between minute 5 and minute 8?

Answer: _____

- (b) What is the rate, in litres per minute?

Answer: _____

6. Two streaming plans charge by the hour.

Plan A is given by the equation $y = 12x$, where y is the cost in dollars for x hours.

Plan B is given by this table.

hours	3	6
cost in dollars	42	84

- (a) What is Plan B's rate, in dollars per hour?

Answer: _____

- (b) Write $<$ or $>$ so the sentence is true:

Plan A's rate _____ Plan B's rate.

Answer: _____

7. Use **Figure 1**. Line **C** passes through $(0, 16)$ and $(8, 0)$. Find its rate of change, and say what the sign of your answer means.

Answer: _____

8. A barrel is leaking. After x minutes it holds $L = -5x + 40$ litres.

(a) What is the rate of change, in litres per minute?

Answer: _____

(b) How many litres are left after 6 minutes?

Answer: _____

9. This table is *not* a straight line, and part (b) is how you can tell.

x	0	2	4
y	0	4	12

(a) Find the rate of change from $x = 0$ to $x = 2$.

Answer: _____

(b) Find the rate of change from $x = 2$ to $x = 4$, then explain how the two answers show the table is not linear.

Answer: _____

10. Mara and Ben both run at a steady speed.

Mara's run is given by this table.

seconds	2	5
metres run	10	25

Ben's run is given by the equation $y = 4.5x$, where y is metres and x is seconds.

(a) What is Mara's speed, in metres per second?

Answer: _____

(b) Write $<$ or $>$ so the sentence is true:

Mara's speed _____ Ben's speed.

Answer: _____

11. Find the slope of the line through $(-3, 5)$ and $(2, -5)$. Take care with the signs: subtract the two y -values and the two x -values in the *same* order.

Answer: _____

12. Synthesis. You have now found four different rates of change on this sheet, one from each kind of representation:

- Line **A** from the graph (Problem 4),
- Line **C** from the graph (Problem 7),
- the tub table (Problem 5),
- the leaking-barrel equation (Problem 8).

List all four rates in order from least to greatest. Then write one sentence saying which representation you found easiest to read a rate from, and why.

Answer: _____